

Multidimensional Balanced Prices [1, 2]

Imagine that a seller has m goods to sell, and n buyers will arrive online one at a time. They have a combinatorial value function $v_i : 2^{[m]} \rightarrow \mathbb{R}_{\geq 0}$ and each function v_i is drawn independently from a known prior distribution. Values and prices are normalized: $v_i(\emptyset) = p(\emptyset) = 0$. Our goal is to set prices such that the welfare of the allocation is maximized.

That is, if we determine the allocation $\mathbf{x}$ such that x_i is the allocation to bidder i , we wish to set a pricing function $p : 2^{[m]} \rightarrow \mathbb{R}_{\geq 0}$ such that under online arrival, the buyers' allocations approximately maximize $\sum_i v_i(x_i)$. Each buyer chooses a utility-maximizing bundle of remaining items at the posted prices.

We will use the notation $\mathbf{x}$ to denote a partial allocation, for instance, the allocation of items after arrival so far. We will use $\mathbf{x}'$ to denote a second allocation that, combined with the partial allocation of $\mathbf{x}$, is still feasible. That is, in this online context, $\mathbf{x}'$ is a feasible allocation of the remaining items not allocated by $\mathbf{x}$, among all buyers. Feasibility means that no item is allocated more than once.

We will use the notation $\text{OPT}(\mathbf{v} \mid \mathbf{x})$ to denote the optimal welfare of all allocations that are feasible with $\mathbf{x}$, i.e., the value remaining (not including the value from $\mathbf{x}$) when $\mathbf{x}$ is already committed to. Formally,

$$\text{OPT}(\mathbf{v} \mid \mathbf{x}) = \max_{\mathbf{x}' : \mathbf{x} \cup \mathbf{x}' \text{ feas.}} \sum_i v_i(x'_i).$$

Definition 1. For $\alpha, \beta > 0$, a pricing rule p is (α, β) -balanced with respect to valuation profile $\mathbf{v} = (v_1, v_2, \dots, v_n)$ if, for all feasible allocations $\mathbf{x}$ and $\mathbf{x}'$ such that $\mathbf{x} \cup \mathbf{x}'$ is feasible:

1. $\sum_i p(x_i) \geq \frac{1}{\alpha} \cdot \underbrace{(\text{OPT}(\mathbf{v}) - \text{OPT}(\mathbf{v} \mid \mathbf{x}))}_{\text{value lost due to allocating } \mathbf{x}}$
2. $\sum_i p(x'_i) \leq \beta \cdot \underbrace{\text{OPT}(\mathbf{v} \mid \mathbf{x})}_{\text{value remaining after allocating } \mathbf{x}}$

Prices are *weakly balanced* if the right-hand side of the second condition is relaxed to $\beta \cdot \text{OPT}(\mathbf{v})$.

We can think of the quantity on the right hand side of (1) as the externality with respect to already allocating $\mathbf{x}$.

Theorem 1. If a pricing rule p is (α, β) -balanced with respect to valuations $\mathbf{v}$, then posting prices $\delta \cdot p$ guarantees value at least $\frac{1}{\alpha\beta+1} \cdot \text{OPT}(\mathbf{v})$ for $\delta = \frac{\alpha}{\alpha\beta+1}$.

Sanity check: In the single-item case, a price $p = \max_i v_i$ is $(1, 1)$ -balanced. Which implies?

A 1/2-approximation of welfare by setting $\frac{1}{2} \max_i v_i$ as the price.

Proof. Let $\mathbf{x}$ be the allocation sold, and let $\mathbf{x}'$ be the allocation achieving $\text{OPT}(\mathbf{v} \mid \mathbf{x})$. Each x'_i was available when buyer i arrived, so their actual utility is at least $v_i(x'_i) - \delta p(x'_i)$. Then we get that

$$\begin{aligned} \text{REVENUE} &= \sum_i \delta \cdot p(x_i) \geq \frac{\delta}{\alpha} (\text{OPT}(\mathbf{v}) - \text{OPT}(\mathbf{v} \mid \mathbf{x})) \\ \text{UTILITY} &\geq \sum_i (v_i(x'_i) - \delta p(x'_i)) \geq (1 - \delta\beta) \text{OPT}(\mathbf{v} \mid \mathbf{x}) \end{aligned}$$

Clearly

$$\text{WELFARE}(\mathbf{x}) = \text{REVENUE} + \text{UTILITY}.$$

Setting δ so that $\frac{\delta}{\alpha} = (1 - \delta\beta)$ concludes the proof. $\square$

Extension to unknown values: For each profile $\mathbf{v}$, take (α, β) -balanced prices $p^\mathbf{v}$, and let $p = \mathbb{E}_{\mathbf{v} \sim \mathbf{F}}[p^\mathbf{v}]$. Then posting prices δp guarantees expected value $\frac{1}{\alpha\beta+1} \cdot \mathbb{E}[\text{OPT}(\mathbf{v})]$. (For weakly balanced prices with $\alpha\beta \geq 1$, use $\delta = 1/(2\beta)$ to obtain $\frac{1}{4\alpha\beta}$.) This extension is stated without proof; see [1].

Construction of Balanced Prices: If each buyer demands at most k items, and if $\mathbf{x}^*$ is an optimal allocation with $|x_i^*| \leq k$, then we set

$$p_j = \begin{cases} \frac{1}{|x_i^*|} v_i(x_i^*) & j \in x_i^* \\ 0 & \text{otherwise} \end{cases}, \quad p(S) = \sum_{j \in S} p_j.$$

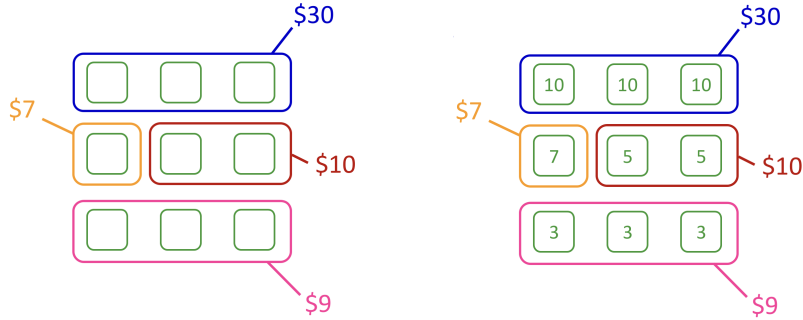


Figure 1: Construction of balanced prices. Figure from Brendan Lucier.

Claim 1. These are weakly $(k, 1)$ -balanced.

Proof. Let $S(\mathbf{x})$ denote the items allocated by $\mathbf{x}$, that is, $S(\mathbf{x}) = \bigcup_i x_i$. Let $I(\mathbf{x}) = \{i : x_i^* \cap S(\mathbf{x}) \neq \emptyset\}$ denote the bidders whose optimal bundles intersect $\mathbf{x}$. For condition (1), we see that

$$\begin{aligned} \forall \mathbf{x}, \quad \sum_i p(x_i) &= \sum_{i: x_i^* \neq \emptyset} \frac{|x_i^* \cap S(\mathbf{x})|}{|x_i^*|} v_i(x_i^*) \\ &\geq \frac{1}{k} \sum_{i \in I(\mathbf{x})} v_i(x_i^*) \\ &\geq \frac{1}{k} (\text{OPT}(\mathbf{v}) - \text{OPT}(\mathbf{v} \mid \mathbf{x})). \end{aligned}$$

The last inequality holds because the untouched optimal bundles remain feasible, so

$$\text{OPT}(\mathbf{v} \mid \mathbf{x}) \geq \sum_{i \notin I(\mathbf{x})} v_i(x_i^*).$$

That is, for any $\mathbf{x}$, the sum of prices of $\mathbf{x}$ is at least $1/k$ of the value of allocations in OPT that intersect $\mathbf{x}$.

For condition (2), we see

$$\forall \mathbf{x}', \quad \sum_i p(x'_i) \leq \sum_{j \notin S(\mathbf{x})} p_j \leq \sum_j p_j = \text{OPT}(\mathbf{v}).$$

Observe that the right hand side does not hold for $\text{OPT}(\mathbf{v} \mid \mathbf{x})$ because item j 's price is the *average* value of the bundle that item j is in in the optimal allocation. However, the value of item j itself may be far lower than the average of the bundle, so this quantity may not be bounded by $\text{OPT}(\mathbf{v} \mid \mathbf{x})$.

That is, after removing $\mathbf{x}$, the total price of the items left over is at most the total price of ALL items, which is $\text{OPT}(\mathbf{v})$.

Our conclusion is then that there exist item prices that guarantee a $4k$ -approx. to the optimal expected welfare, by posting half of each item's expected constructed price. $\square$

Additional Resources

I'm not aware of any textbooks on the subject, but here is a list of resources on prophet inequalities and the breadth of work in more recent research:

- 2017 Survey “An Economic View of Prophet Inequalities” by Brendan Lucier [3]:
https://sigecom.org/exchanges/volume_16/1/LUCIER.pdf
- 2016 Simons Bootcamp Talks by Matt Weinberg
 - Part I: <https://www.youtube.com/watch?v=NwF4Xr0-6Rc>
 - Part II: <https://www.youtube.com/watch?v=E19TWolv8I>
- EC 2021 Tutorial on Prophet Inequalities by Michal Feldman, Thomas Kesselheim, and Sahil Singla
 - Website (slides and reading list): <http://www.thomas-kesselheim.de/tutorial-prophet-inequalities/>
 - Part 1: <https://www.youtube.com/watch?v=qbHd0g9RkCg>
 - Part 2: <https://www.youtube.com/watch?v=l20KP5IIgcQ>
 - Part 3: <https://www.youtube.com/watch?v=lyOUcYfNEiA>

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References

- [1] Paul Dütting, Michal Feldman, Thomas Kesselheim, and Brendan Lucier. Prophet inequalities made easy: Stochastic optimization by pricing non-stochastic inputs. In Chris Umans, editor, *58th IEEE Annual Symposium on Foundations of Computer Science, FOCS 2017, Berkeley, CA, USA, October 15-17, 2017*, pages 540–551. IEEE Computer Society, 2017.
- [2] Michal Feldman, Nick Gravin, and Brendan Lucier. Combinatorial auctions via posted prices. In *Proceedings of the Twenty-Sixth Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 123–135. SIAM, 2015.
- [3] Brendan Lucier. An economic view of prophet inequalities. *SIGecom Exch.*, 16(1):24–47, September 2017.
- [4] Tim Roughgarden. *Twenty Lectures on Algorithmic Game Theory*. Cambridge University Press, 2016.